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Professional Profile

I am an M.S. candidate in the AI-Embedded Software on Chip Lab at Kyungpook National University, advised by Prof. Daejin Park. I work on **real-time embedded systems and timing-predictable automotive software for Software-Defined Vehicles (SDVs)**. I take control and AI algorithms past the model stage onto automotive MCUs (Infineon AURIX TC375/TC397, NXP S32K344), real motors, in-vehicle networks, and embedded Linux, working down to linker scripts, bootloaders, and device drivers, and measure how they actually behave. My work includes an end-to-end HILS demonstrator for a **Hyundai Motor Company** project, first-author papers at **EMSOFT 2026** (to appear in **IEEE TCAD**) and in **IEEE Access**, first-author presentations at seven international conferences (six oral, one poster), and talks at lab tutorial seminars, the lab symposium, and industry workshops.

Research Interests: Real-Time Embedded Systems · Timing-Predictable Automotive Software · Software-Defined Vehicles · EIL/HIL Validation · Mixed-Criticality Control · HW/SW Co-Design · Embedded AI

EMSOFT'26 / IEEE TCAD | **IEEE Access** | **Hyundai Motor HILS** | **6 Oral + 1 Poster** | **7 SW Copyrights** | **13 Lab Talks**

Education

Kyungpook National University (KNU), Daegu, Korea — Mar 2025 – Feb 2027 (expected)
M.S. in Electronic and Electrical Engineering, School of Electronic and Electrical Engineering — GPA 4.22/4.30
Convergence Major in Future Vehicle Software | Advisor: Prof. Daejin Park | AI-Embedded Software on Chip Lab
Thesis (in progress): *On the Hardware–Software Gap in Runtime Control* (working title)
Coursework: Advanced Operating Systems, Advanced Computer Architecture, Embedded Software Design, Special Topics in Control II, Future Vehicle SW Convergence Seminar I, Industry–Academia Projects

Kyungpook National University (KNU), Daegu, Korea — Mar 2019 – Feb 2025
B.S. in Electronics Engineering, School of Electronics Engineering, College of IT Engineering — GPA 3.72/4.30
Undergraduate researcher, AI-Embedded Software on Chip Lab (2024 – 2025)
Coursework: Operating Systems, Computer Architecture, Microprocessor Design Lab, Data Structures, Automatic Control

Industry–Academia Project

Real-Time HILS for CNN-Based Motor Fault Diagnosis — Mar 2026 – Oct 2026
Hyundai Motor Co. — *PHM Fault-Diagnosis Model Lightweighting & Embedded Methodology*; HILS architecture & integration

- Designed and built the end-to-end HILS: **CARLA vehicle speed drives a real PMSM (NXP S32K344), the teammates' CNN on an AURIX TC397 diagnostic ECU classifies motor signals, and the result feeds back into motor control and CARLA**.
- The model was trained on 25 kHz accelerometer and flux-sensor signals that the rig could not provide, so built vibration/flux proxy signals from motor-control internals in 512 ms windows, sent over a CAN protocol that never disturbs control frames and drops incomplete windows.
- Moved feature extraction onto the TC397 as a streaming Hann + Goertzel stage computing only the 22 required frequencies, and reproduced all five diagnosis classes by injecting fault signatures at the measured-signal stage.
- Added a link-loss fail-safe (0.5 s without CARLA input → self-generated zero-speed command and normal stop; detected in 0.54 s) and a Hyundai-requested fault reaction (three consecutive decisions at $\geq$70% confidence slow or stop the motor and the CARLA vehicle with a warning). Raised the reply rate from 83% to 100% by tracing 8 ms RX-buffer overflows to blocking logging and built-in self-test inference.
- In a 15-minute drive, there were no false decisions without faults, and all 12 injected faults (4 types $\times$ 3) were confirmed. Proxy signals differ from the training data, so results cover the real-time pipeline, not diagnostic accuracy.
- Live-demonstrated it to Hyundai after the final report at the Namyang R&D Center (Oct 16, 2026) on two pre-configured laptops, with a setup manual for Hyundai to rebuild the environment.

Research Experience

On the Hardware–Software Gap in Runtime Control (working title) Aug 2026 – Present
M.S. Thesis — binding-time analysis on Xilinx Zynq-7020 and NXP S32K344

- Read the hardware–software gap in runtime control as a gap in *binding time*, the instant a result becomes fixed, and root-caused fragmented control performance at clock-cycle and register level with paired HW/SW implementations: a commit 160 ns later separated a stable loop from an unstable one, and the *order* of software delays decided stability.
- Ran MLPerf Tiny inference beside the control loop on one core, across two cores, and on an FPGA accelerator: kernels 1.9–27× faster delayed the control commit further, and the same model in logic moved its result time by at most 0.51% versus up to 11.8% in software. On an S32K344 driving a real motor, binding the commit to the PWM absorbed the inference delay.
- Proposed a *binding region* that isolates the binding rather than the task (about 3% of the logic). It committed beside every AI workload as if alone, earlier than literature remedies on the same device, at no cost to the AI and across six control laws, and held while a runtime-growing AI swapped in new models up to 264 KB. 35 pre-registered measurements; 193 of 205 hypotheses held.

Command-Authority Scheduling for Mixed-Criticality SDV Control May 2026 – Oct 2026
[J4] IEEE Access (early access, 2026)

- Proposed CAS, which converts low-criticality service outputs into role-constrained authority claims under per-tick ceilings synthesized from an identified plant model and an ISO 15622 / RSS / UN R79 response envelope.
- On a FreeRTOS-based AURIX TC375 ECU-in-the-loop platform with real-time CARLA, CAS was the only service-integrating policy that stayed inside the envelope in all runs. Against a service-free baseline it cut RSS-distance violations by 97% and safety-governor interventions by 88% at a 1.2% lower mean speed, with a worst-case receive-to-transmit time of 171 μs (1.7% of the 10 ms period, debug build).

Reproducible Distributed ECU-in-the-Loop Co-Simulation (JSEC) Oct 2025 – Aug 2026
[J3] EMSOFT 2026 / IEEE TCAD

- Built a distributed CARLA–Simulink (14-DOF vehicle dynamics)–AURIX TC375 testbed over UDP, with a byte-level TRIG/ACT/LOG packet protocol and non-blocking logging.
- Proposed JSEC, which maps admitted ECU events to fixed-step plant advancement with bounded catch-up execution. Raised mean R^2 across repeated runs from 0.410 to 0.947 with 0.79–1.14% execution-time overhead.

On-Chip ADC Variability for Early Detection of Software-Induced Margin Degradation Dec 2025 – Jun 2026
[C6] IPFA 2026

- Designed five software-induced stress scenarios on an AURIX TC375 and showed that stress first appears as increased ADC output *variability* while ISR runtime and ADC means stay nominal.
- Defined a 3σ threshold that separates latent stress from environmental noise, with no extra sensors or hardware changes.

Architecture-Aware Neural Firmware Compression via Knowledge Distillation Jun 2025 – Jan 2026
[C5] ICAIIC 2026

- Distilled a Transformer teacher into a lightweight GRU student that predicts the next byte of TriCore firmware and drives an arithmetic coder, trained on about 60 TC375 firmware images.
- Compressed firmware to 19.05% of its original size, about 38% better than LZMA, with a 0.25 MB decoding footprint (about 94% lower). Python-vs-C cross-validation confirmed bit-exact decoding.

Earlier Research: Embedded Vehicle Control, Perception, and Emulation Jan 2024 – Sep 2025
[C1]–[C4], [J1], [J2]

- **Adaptive sliding-mode control** [C4]: tuned Super-Twisting gains online with a rule-based empirical RL update (no neural-network training) and blended linear and nonlinear sliding surfaces.
- **Multithreaded PID control** [C3]: parallelized sensing, computation, and actuation with precomputed control candidates in portable C on QEMU–Simulink, cutting average settling steps by about 32.3%.
- **Camera-only depth mapping** [C2, J2]: LiDAR-free stereo depth maps with a custom 3×3 convolution kernel and shadow removal, at real-time rates without deep-learning models.
- **ADAS emulation and adaptive parking** [C1, J1]: Smart Cruise Control and Lane Keeping Assist with YOLO detection and ESP32 control, plus a driver-adaptive four-camera parking assistant.

Engineering Projects

TC375 Bootloader and Linker-Script Partitioning Aug 2025
TASKING linker script (LSL), startup code, boot flow

- Analyzed the TC375 TASKING linker script and split it into a bootloader image and an application image; prototyped a minimal bootloader that checks the application entry, disables interrupts, and jumps to the application.

TriCore Firmware Binary Analysis and Runtime Code Loading Aug 2025

Linker maps, Intel HEX, instruction encoding

- Patched variable initial values and retargeted a function call directly in Intel HEX without rebuilding, and verified the change on the board.
- XOR-encrypted a function body in Flash, then decrypted and copied it into RAM at runtime and executed it through a function pointer (a TRAP occurs without decryption).

Simulink MBD-to-AURIX Code Integration Automation Jul 2025

Model-based design, code generation, Python tooling

- Wrote Python tools that parse Embedded Coder output by function-name and comment patterns, insert TC375 PWM/ADC driver code, and generate the main source file, replacing manual integration.
- Validated Sine→PWM and ADC→PID→PWM models end to end on a real AURIX TC375 board.

Linux UART Device Driver with CRC/ECC and Huffman Coding Dec 2024 – Jan 2025

Lab coding camp — team lead (5 members), integration & debugging

- Wrote a GPIO-based UART driver for two BeagleBone boards (Debian, kernel 4.19) as a loadable kernel module exposing a character device: a falling-edge IRQ starts an `hrtimer` that samples each bit and re-arms with `hrtimer_forward_now()`, with the sender and receiver timers aligned to the same bit time.
- Appended SEC-DED (7 bit) and CRC16 to each 32-bit block; the receiver runs SEC-DED only when CRC fails, and a receive log showed a CRC-failed block recovered. With Huffman coding, 1-bit errors were detected but not corrected (variable-length bits broke the fixed-block checks); identified per-block CRC on the compressed stream as the remedy (not implemented) and built a double-buffered encode/transmit pipeline (two threads, mutex, condition variable), leaving UART integration and performance comparison as future work.

uClinux Bring-Up on STM32F429 Nov 2024

Embedded Linux, bootloader, toolchain

- Built and flashed U-Boot, an XIP kernel, and ROMFS on an STM32F429I-DISC1, fixing host toolchain issues, ran cross-compiled BFLT applications from a UART console, and traced the VMA/LMA layout and the BFLT loader's Flash-to-RAM relocation on a no-MMU system.

Adaptive Auto-Parking Control: PID, MPC, and Disturbance Compensation Aug 2024 – Sep 2024

Control design in MATLAB/Simulink and C

- Implemented gain-scaled adaptive PID (Simulink) and MPC (MATLAB and C), and added Kalman-filtered disturbance compensation that suppresses control spikes at disturbance onset.

Teaching & Technical Seminars

Guest Lecturer & Lab Assistant, LIG Nex1 Embedded Systems Workshop Sep 2025

Industry workshop organized by Prof. Daejin Park

- Delivered the uClinux session (building uClinux on STM32F429, explaining XIP execution and BFLT relocation with `objdump`) and ran the TRACE32 hands-on demo on a TC375 (breakpoints and peripheral views).

Teaching Assistant, Embedded Software Lectures for Industry Engineers Multiple sessions

Hyundai Motor Company, Hyundai AutoEver, LIG Nex1 (basic/advanced) — lab setup and hands-on support

Lab Tutorial Seminars & Symposium Talks Jul 2024 – Sep 2025

AI-Embedded Software on Chip Lab — 13 hands-on presentations with board results and troubleshooting notes

- **MCU firmware & debugging:** TRACE32 (ARM & TriCore), Infineon MemTool, TriCore HEX modification, Simulink MBD on TriCore, NXP S32K344 PMSM FOC, STM32–Simulink.
- **Embedded Linux & virtualization:** uClinux on STM32F429 (U-Boot, XIP kernel, BFLT loader), QEMU/KVM setup, Simulink–QEMU (Cortex-M3 ELF, startup code, linker script).
- **Control & simulation:** MPC in MATLAB and C, linking MATLAB & C (MEX, Engine API, Coder), reinforcement learning in MATLAB, CARLA simulator.

Publications

IEEE TCAD (EMSOFT) ×1 **IEEE Access ×1** **KCI Journal ×2** **Int'l Conference ×6** First author on all 10

Journal Articles

[J4] **H. Lee** and D. Park, "Command-Authority Scheduling for Bounded Low-Criticality Service Influence in Mixed-Criticality SDV Control," *IEEE Access*, early access, 2026, doi: 10.1109/ACCESS.2026.3749257.

[J3] **H. Lee** and D. Park, "JSEC: Jitter-to-Step Execution Contracts for Reproducible Distributed ECU-in-the-Loop Co-Simulation," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, to appear, doi: 10.1109/TCAD.2026.3722824.
EMSOFT 2026 full paper, journal track (ACM/IEEE Int. Conf. on Embedded Software), accepted Aug. 2026; presented at ESWEEK 2026, Barcelona, Spain, Oct. 7, 2026. (Oral)

[J2] **H. Lee** and D. Park, "Real-Time Depth Map Using Stereo Vision and Convolutional Filtering," *Journal of the Korea Institute of Information and Communication Engineering (JKIICE)*, vol. 29, no. 9, Sep. 2025. (KCI)

[J1] **H. Lee** and D. Park, "Auto Parking Assistance Control using Human-Activity Feedback-based Embedded Software Emulation," *Journal of the Korea Institute of Information and Communication Engineering (JKIICE)*, vol. 28, 2024. (KCI)

International Conference Papers

[C6] **H. Lee** and D. Park, "Early Detection of Software-Induced Internal Margin Degradation in Automotive MCUs Using On-Chip ADC Variability," in *Proc. IEEE Int. Symp. on the Physical and Failure Analysis of Integrated Circuits (IPFA)*, Singapore, Jul. 2026. (Oral)

[C5] **H. Lee**, H. Kim, and D. Park, "Architecture-Aware Neural Compression for TriCore Firmware using Knowledge Distillation," in *Proc. Int. Conf. on Artificial Intelligence in Information and Communication (ICAIIC)*, Tokyo, Japan, Feb. 2026. (Oral)

[C4] **H. Lee** and D. Park, "Lightweight Empirical Reinforcement Learning Driven Adaptive Super-Twisting Control with Fused Linear-Nonlinear Sliding Surfaces for Embedded Vehicle Control," in *Proc. 51st Annual Conf. of the IEEE Industrial Electronics Society (IECON)*, Madrid, Spain, Oct. 2025. (Oral)

[C3] **H. Lee** and D. Park, "Advanced Real-Time Performance QEMU Emulated Automotive Control System with a Multithreaded PID Controller," in *Proc. IEEE Int. Conf. on Consumer Electronics – Taiwan (ICCE-TW)*, Kaohsiung, Taiwan, Jul. 2025. (Oral)

[C2] **H. Lee** and D. Park, "Convolution-Based Depth Map With Shadow Removal Using Cameras for 3D Mapping in Autonomous Vehicle Driving," in *Proc. IEEE Int. Symp. on Intelligent Signal Processing and Communication Systems (ISPACS)*, Kaohsiung, Taiwan, Dec. 2024. (Oral)

[C1] **H. Lee** and D. Park, "Multi-Camera Interoperable Emulation Framework Using Embedded Edge-Cloud AI Computing for Autonomous Vehicle Driving," in *Proc. IEEE Vehicular Networking Conf. (VNC)*, Kobe, Japan, May 2024. (Poster)

Software Copyright Registrations

Korea Copyright Commission; holder: KNU Industry–Academic Cooperation Foundation; co-developed with Prof. D. Park.

[S7]	TriCore-Based Intel HEX Firmware Compression, No. C-2025-044871	Oct 2025 · [C5]
[S6]	Real-Time Depth Map Using Stereo Vision, No. C-2025-044868	Oct 2025 · [J2]
[S5]	Adaptive Super-Twisting Control, No. C-2025-044867	Oct 2025 · [C4]
[S4]	QEMU-Based Multi-Threaded Automotive Control, No. C-2025-044866	Oct 2025 · [C3]
[S3]	Depth Map Generation Program for 3D Mapping, No. C-2024-054193	Dec 2024 · [C2]
[S2]	Autonomous Driving Emulation Program, No. C-2024-052879	Dec 2024 · [C1]
[S1]	Parking Assistance Control Adjustment, No. C-2024-051914	Dec 2024 · [J1]

Technical Skills

Programming	C, C++, Python, MATLAB, Bash; reading ARM and TriCore assembly; Makefiles and linker scripts
MCUs & Boards	AURIX TC375/TC397 (TriCore), NXP S32K344, STM32F429, Xilinx Zynq-7020, BeagleBone
Firmware & OS	Bootloaders, linker scripts (TASKING LSL, GNU ld), startup code, Timers/ISR, PWM, ADC, DMA, FreeRTOS, uClinux, Linux kernel-module device drivers, cross-compilation, ELF/map/Intel HEX
Networks	CAN, UART, UDP, TCP/IP, Ethernet (LwIP)
Debug & Flash	TRACE32, Infineon MemTool, FreeMASTER, AURIX Dev. Studio, S32DS, EB tresos, OpenOCD
Simulation	MATLAB/Simulink, Embedded Coder, CARLA, QEMU/KVM, ROS/ROS 2 (CARLA bridge setup)
Control	PID, MPC, sliding-mode/super-twisting control, PMSM FOC, Kalman filter, disturbance observer
AI/ML & Vision	PyTorch, RL (DQN, PPO), imitation learning, knowledge distillation, OpenCV, YOLO, Open3D

Languages, Certifications & Other

- **Languages:** Korean (native), English (OPIc Intermediate High, ACTFL) — May 2026
- **Computer Specialist in Spreadsheet & Database, Level 1**, Korean national certification (KCCI) — Oct 2020
- **XR Expert Training Program** (48 h, Unity/C#), Daegu VR·AR Production Center — Oct 2022 – Dec 2022
- **Military Service:** completed as Social Service Personnel — Jan 2021 – Oct 2022

References

Prof. Daejin Park (M.S. Advisor), School of Electronics Engineering, KNU — boltanut@knu.ac.kr